



Inspiring Excellence

CSE 251

Electronic Devices and Circuits

Lecture 6

Course instructor:

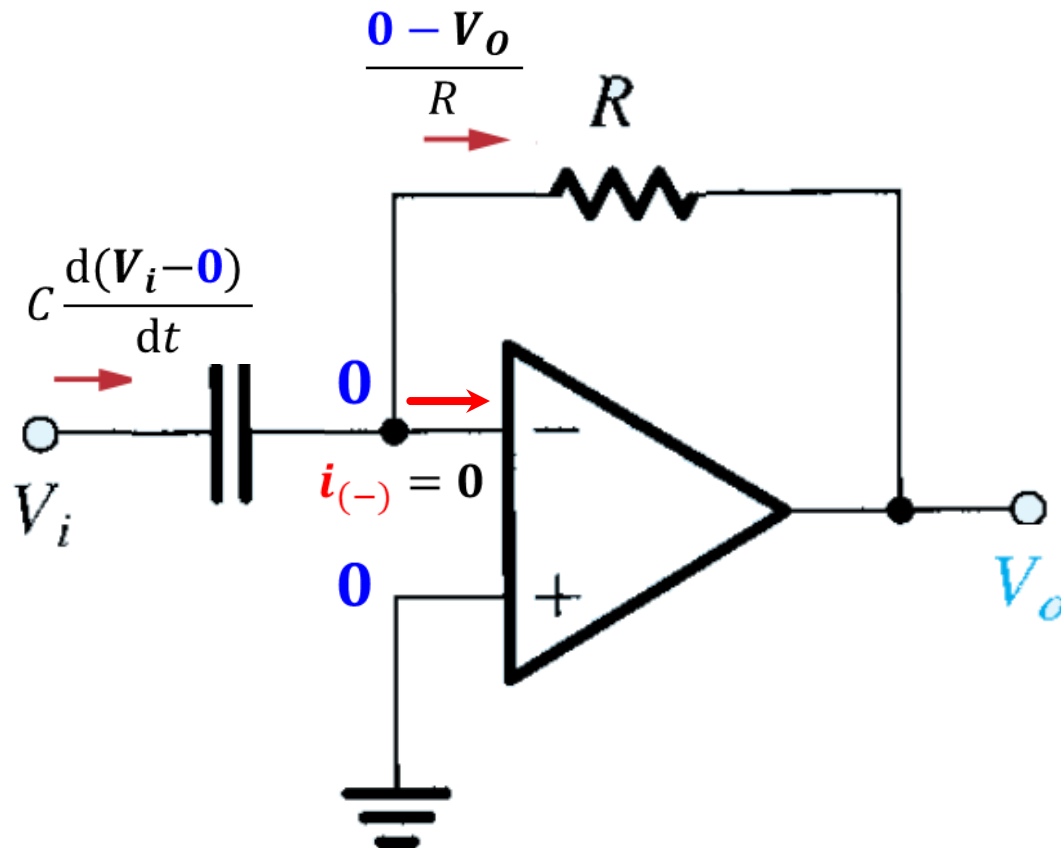
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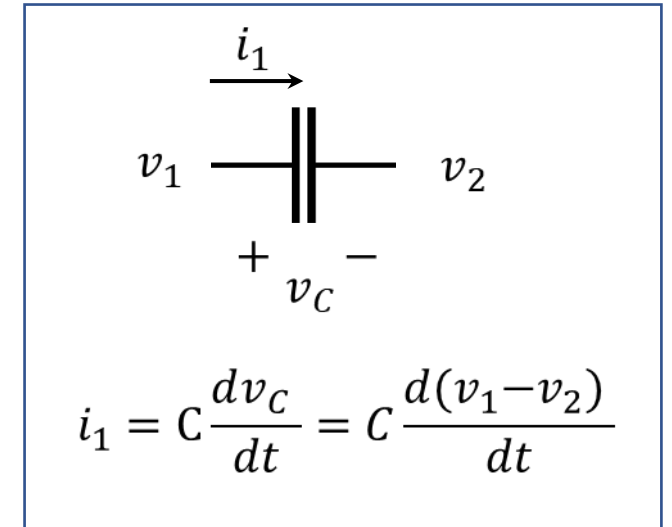
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Op Amp as Inverting Differentiator

Since ideal op-amp, $i_- = i_+ = 0$, so $i_1 = i_2$



Review – Capacitor

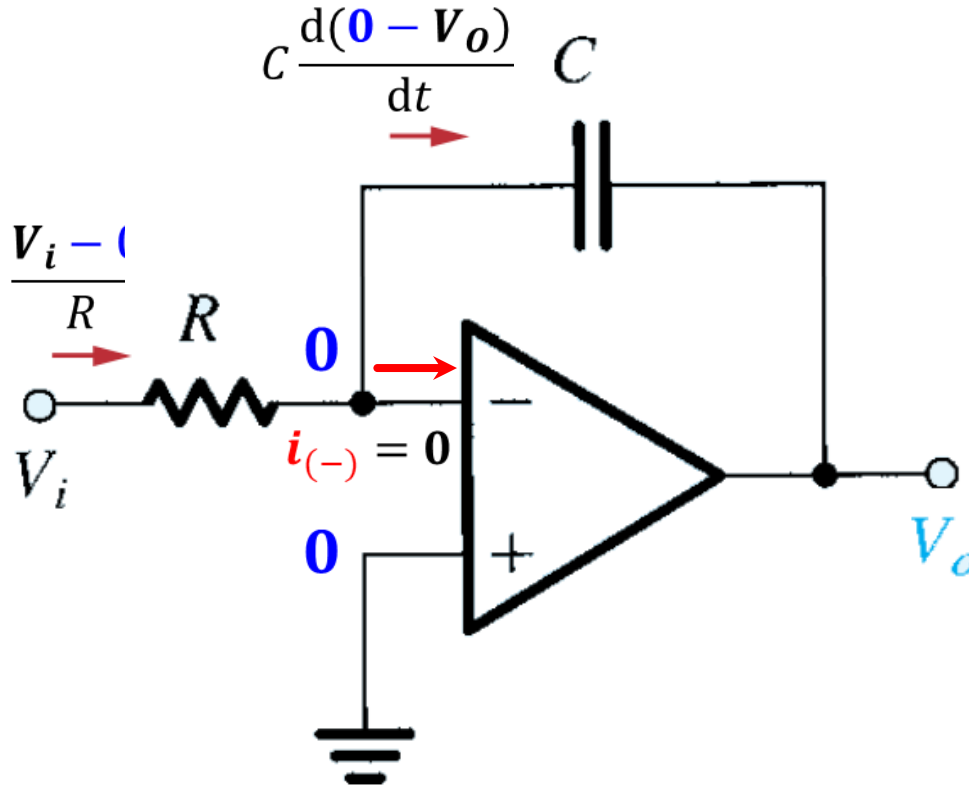


$$\Rightarrow -\frac{V_o}{R} = C \frac{dV_i}{dt}$$

$$\Rightarrow V_o = -RC \frac{dv_i}{dt}$$

Op Amp as Inverting Integrator

Since ideal op-amp, $i_- = i_+ = 0$, so $i_1 = i_2$



Review – Capacitor

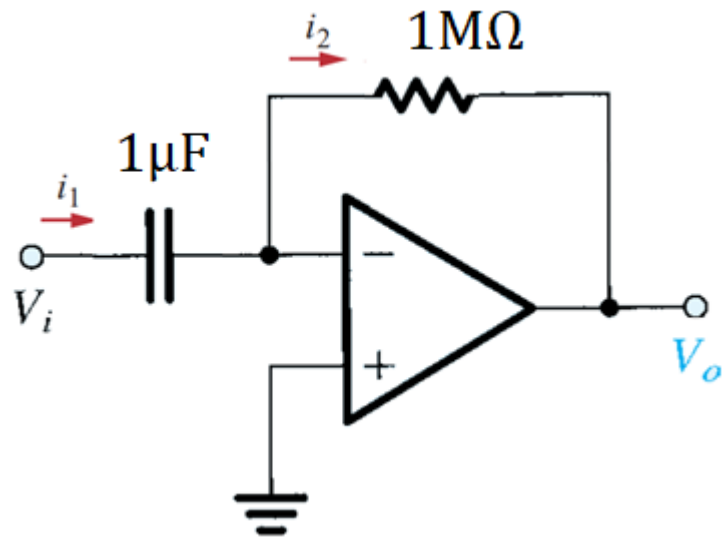
$$\begin{array}{c} i_1 \rightarrow \\ v_1 \text{ --- } \parallel \text{ --- } v_2 \\ \quad + \quad v_C \quad - \\ i_1 = C \frac{dv_C}{dt} = C \frac{d(v_1 - v_2)}{dt} \end{array}$$

$$\Rightarrow \frac{V_i}{R} = -C \frac{dV_o}{dt}$$

$$\Rightarrow V_o = -\frac{1}{RC} \int V_i(t) dt$$

Example 10

Observe the following Figure. If $V_i = 5 \cdot \sin(6t)$, Find the value of V_o



Solution:

This is a **differentiator**.

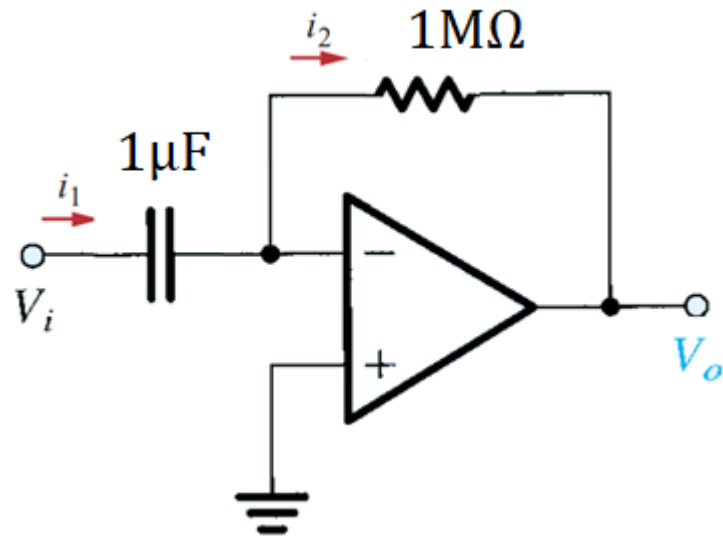
$$v_o = -RC \frac{dV_i}{dt}$$

$$= -(1 \times 10^6) \cdot (1 \times 10^{-6}) \times \frac{d(5 \cdot \sin(6t))}{dt}$$

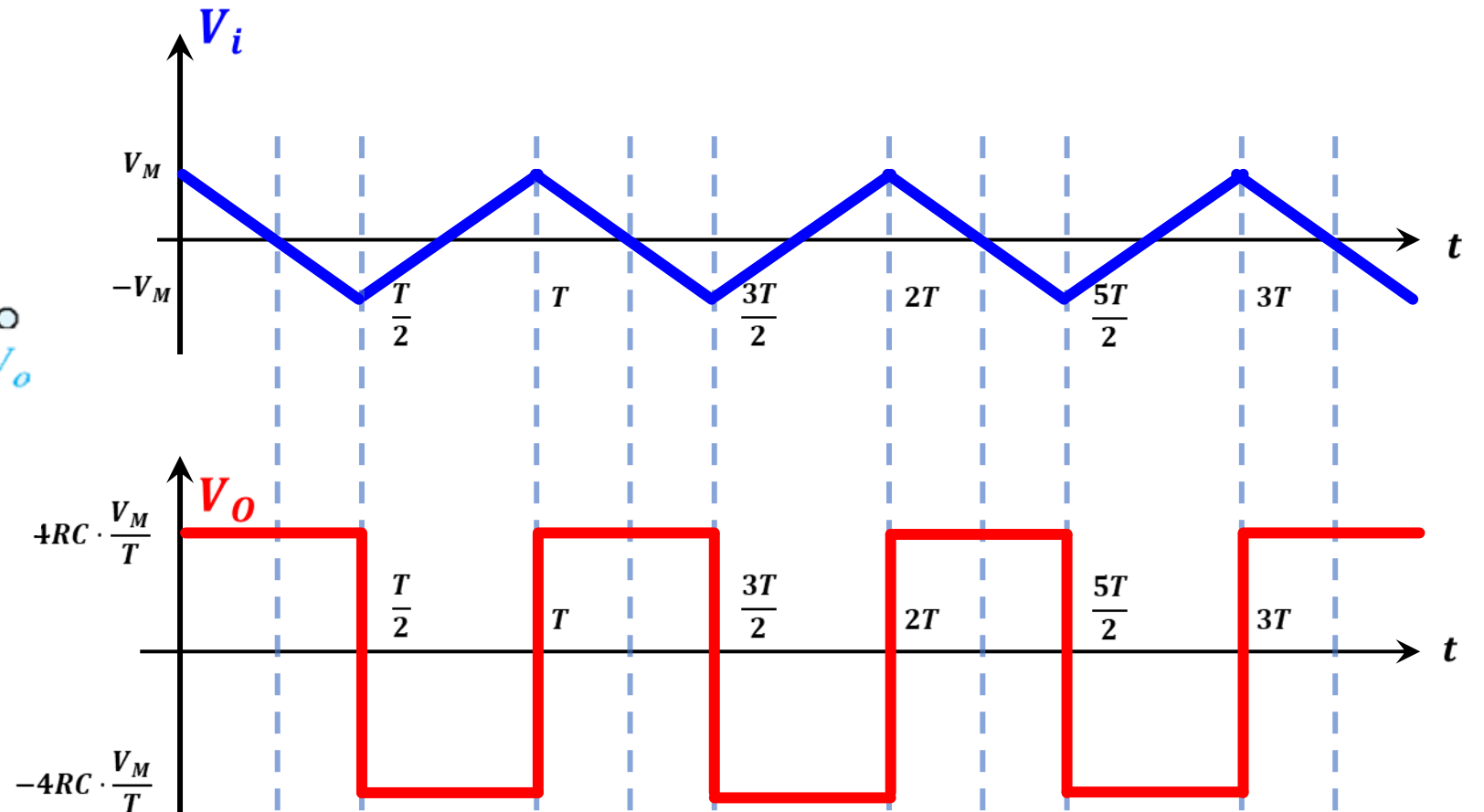
$$= -1 \times (5 \times 6 \cos(6t))$$

$$= \mathbf{-30 \cos(6t)}$$

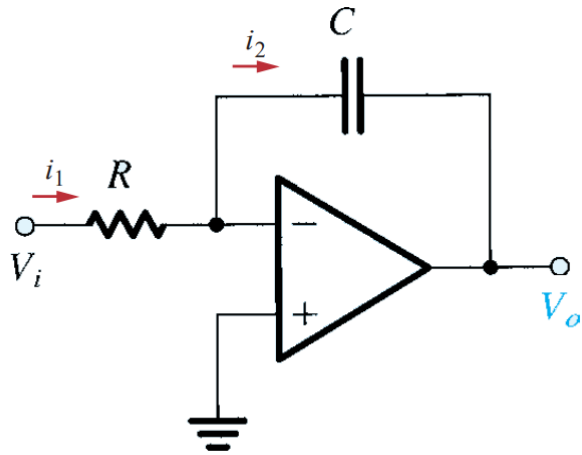
Example 11



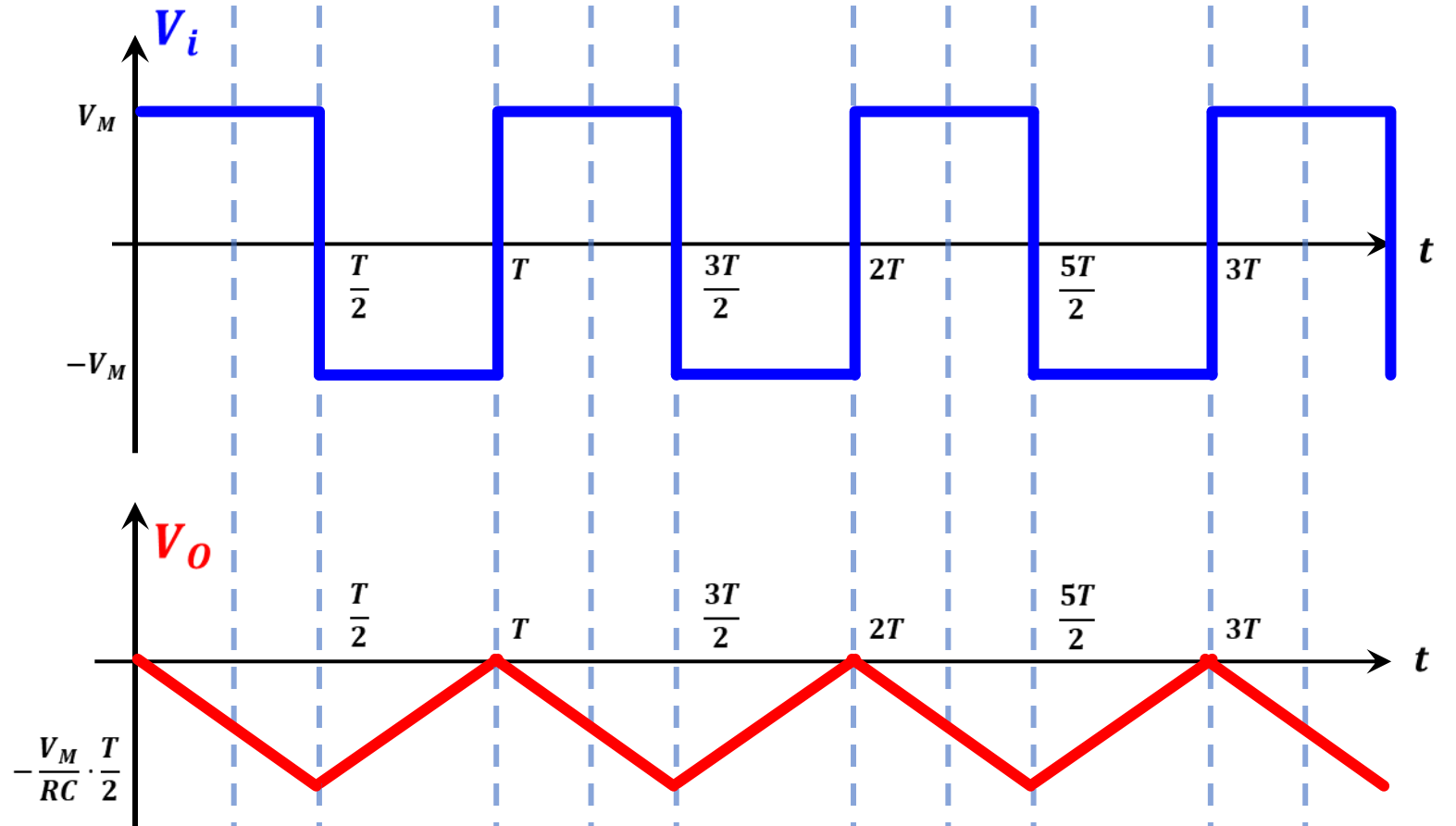
$$\text{Slope: } \left| \frac{dv}{dt} \right| = \frac{V_M - (-V_M)}{T/2} = \frac{4V_M}{T}$$



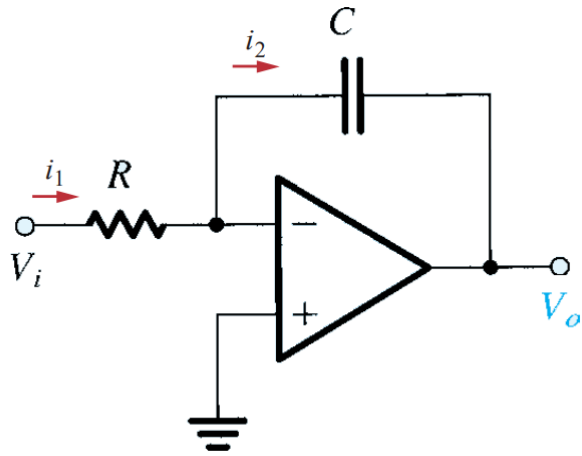
Example 12



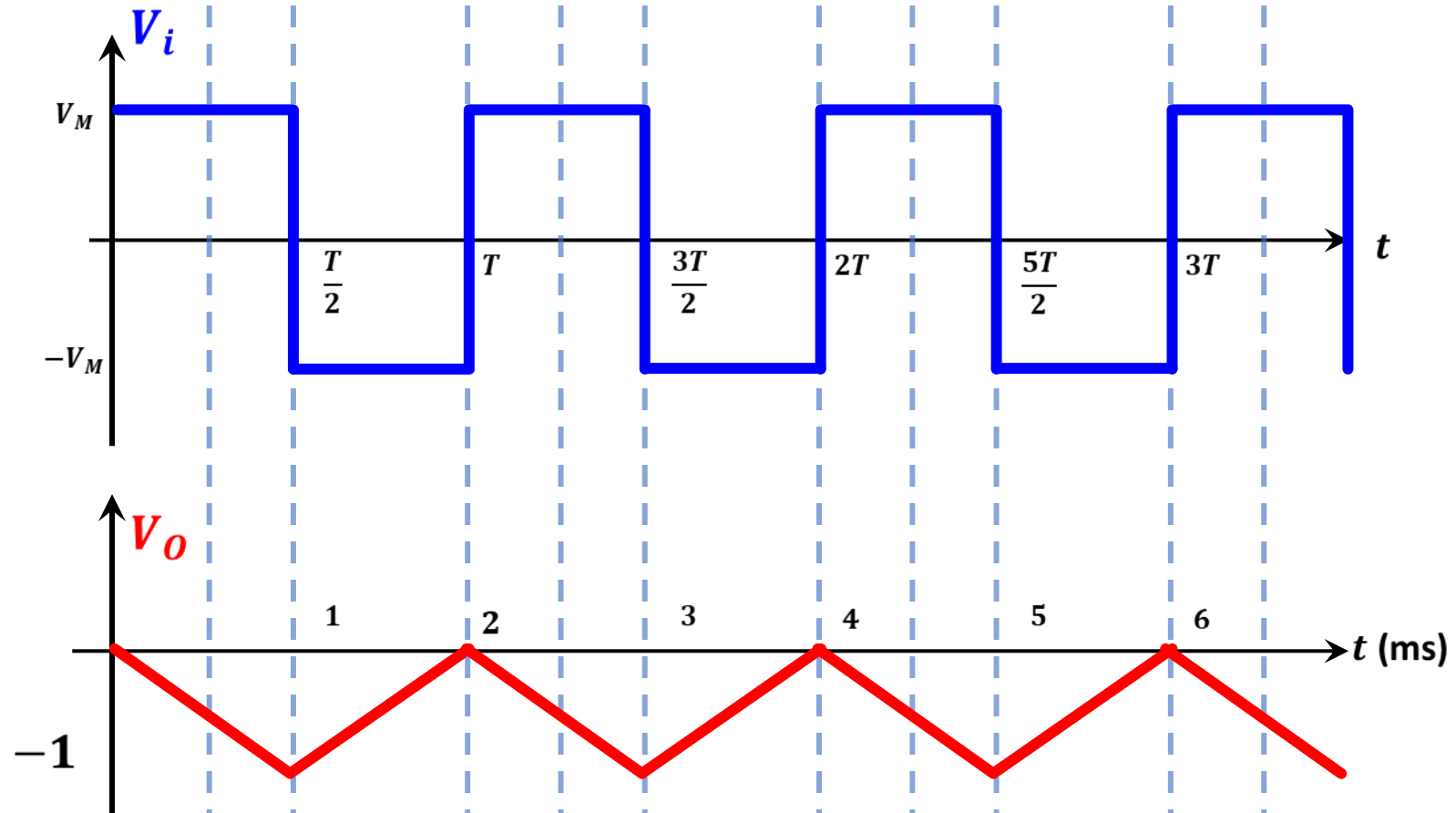
$$\int_0^t v_I dt = V_M \cdot t$$



Example 12



$$\int_0^t v_I dt = V_M \cdot t$$



APPLICATIONS:

Implementing operational functions

- $f = -2x - 3y$
- $f = -4x + 5y$
- $f = -7x + \frac{d}{dt}y$
- $f = -3\frac{d}{dt}z + 6x + 9\int y \cdot dt$
- $f = \frac{d^2x}{dt} + 10y + \int(10z - 9)dt$
- $f = -\frac{1}{3}\int x \cdot dt + 2 \ln y + 4z$
- $f = -3\frac{dx}{dt} + 2 \exp(y) + 4z$
- $f = xy/z$

APPLICATIONS:

implementing operational functions

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implementing operational functions

- $f = -4x + 5y$

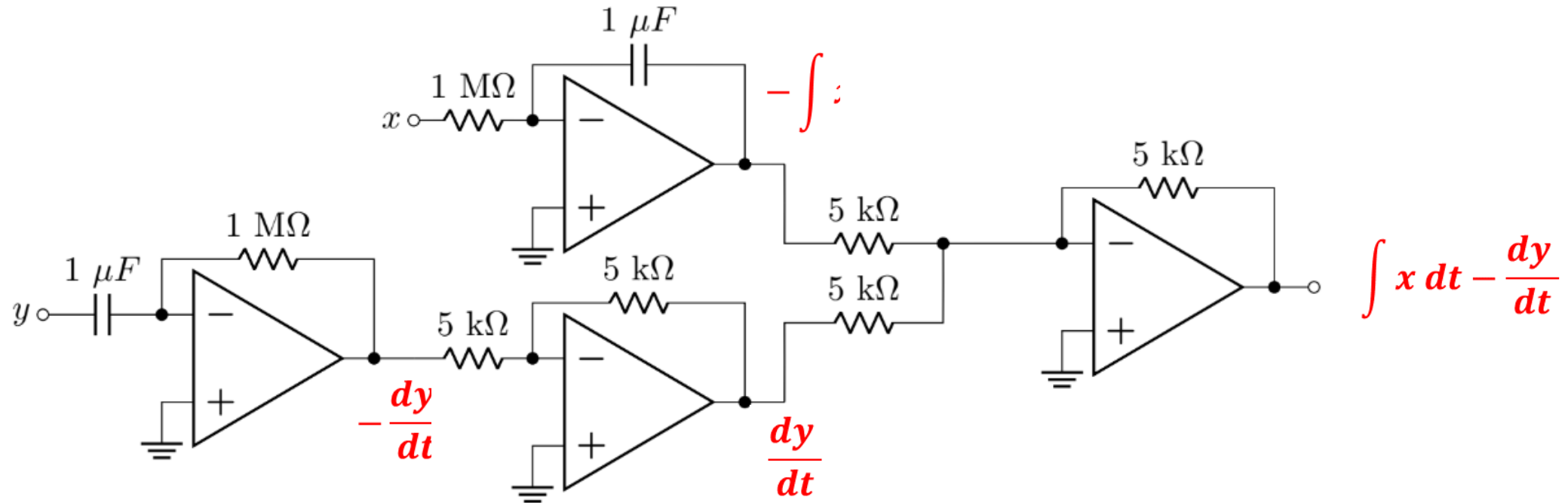
APPLICATIONS:

implementing operational functions

- $f = -7x + \frac{d}{dt}y$

Example

Analyze the circuit below to **find** an expression of **f** in terms of inputs **x** and **y** .



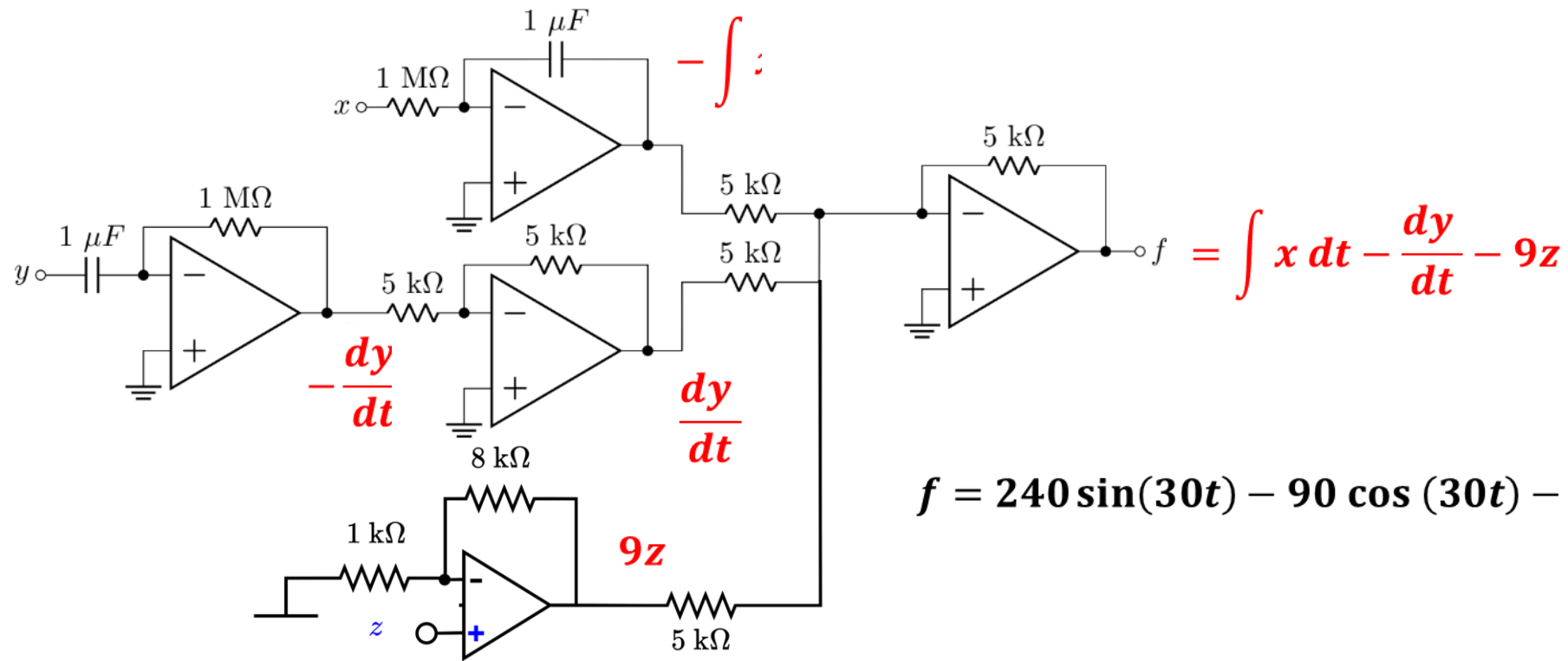
Solution:

$$v_{I_1} = -\frac{dy}{dt}; v_{I_2} = -\frac{1}{RC} \int x\ dt; v_{I_3} = -v_{I_1} = \frac{dy}{dt}; v_o = -(v_{I_2} + v_{I_3})$$

Example

Analyze the circuit below to **find** an expression of **f** in terms of inputs **x** and **y** .

If **$x = 8 \cos(30t)$ V** and **$y = 3 \sin(30t)$ V** and **$z = 1$ V**

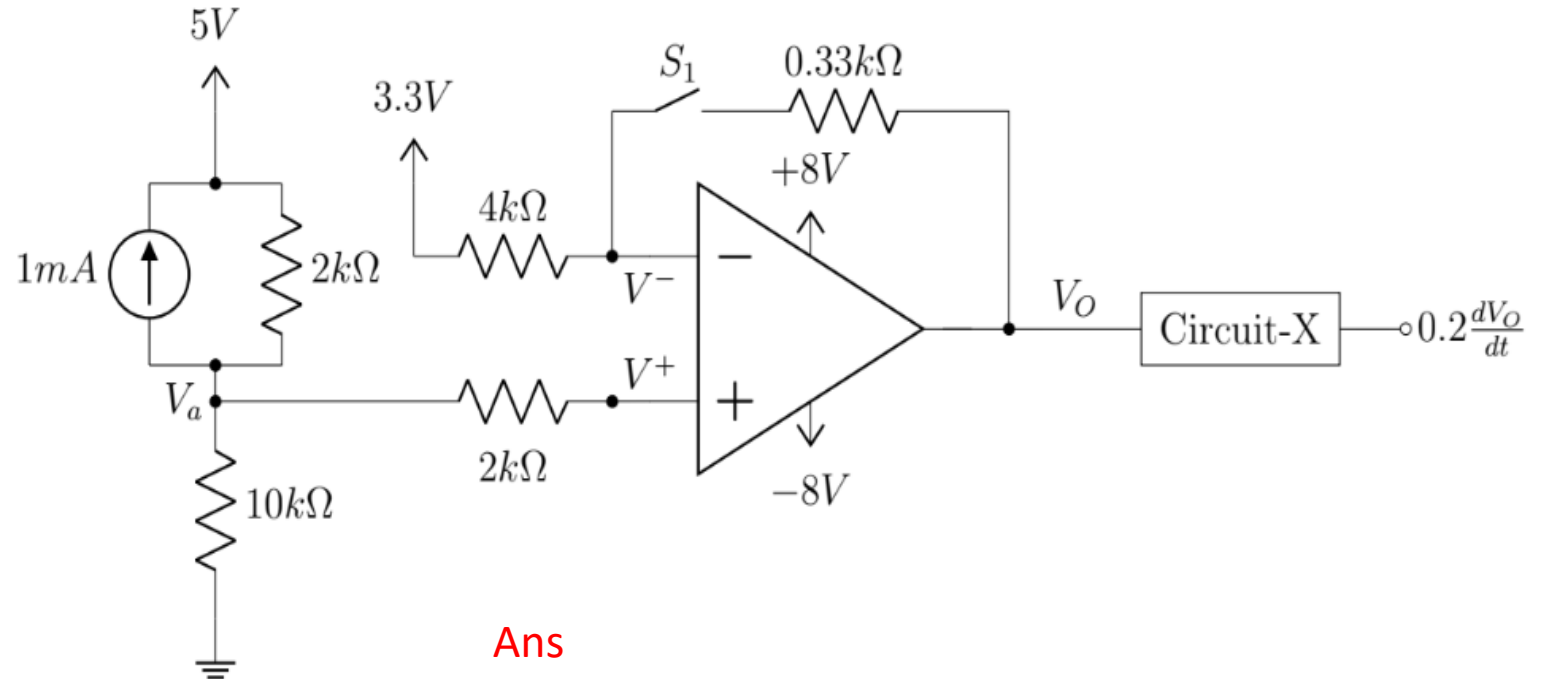


$$f = 240 \sin(30t) - 90 \cos(30t) - 9 \text{ V}$$

Problem 1

The circuit diagram has a switch S_1 which is shown to be 'open' in the figure. The output V_O is passed through an unknown block of 'Circuit-X' and a differentiated result is generated.

- a) **State** the equation of gain of an inverting amplifier.
- b) **Calculate** the Value of V_a and V_+ .
- c) **Determine** V_O when switch S_1 is closed
- d) **Determine** V_O when switch S_1 is open
- e) **Design** the circuit-X. Assume any value if necessary.



Ans

b) 2.5v

c) 2.434

d) -8v

Thank You!